

Using the Fractal Drawing Function

Now that we have two sample drawing functions to work with, let's examine the main function in the Fractal Art Maker project, `drawFractal()`. This function has three required parameters and one optional one:

`shapeDrawFunction`, `size`, `specs`, and `maxDepth`.

The `shapeDrawFunction` parameter expects a function, like `drawFilledSquare()` or `drawTriangleOutline()`. The `size` parameter expects the starting size passed to the drawing function. Often, a value between 100 and 500 is a good starting size, though this depends on the code in your shape-drawing function, and finding the right value may require experimentation.

The `specs` parameter expects a list of dictionaries that specify how the recursive shapes should change their size, position, and angle as `drawFractal()` recursively calls itself. These specifications are described later in this section.

To prevent `drawFractal()` from recursing until it causes a stack overflow, the `maxDepth` parameter holds the number of times `drawFractal()` should recursively call itself. By default, `maxDepth` has a value of 8, but you can provide a different value if you want more or fewer recursive shapes.

A fifth parameter, `depth`, is handled by `drawFractal()`'s recursive call to itself and defaults to 0. You don't need to specify it when you call `drawFractal()`.

Setting Up the Function

The first thing the `drawFractal()` function does is check for its two base cases:

Python

```
def drawFractal(shapeDrawFunction, size, specs, maxDepth=8, depth=0):  
    if depth > maxDepth or size < 1:  
        return # BASE CASE
```

If `depth` is greater than `maxDepth`, the function will stop the recursion and return. The other base case occurs if `size` is less than 1, at which point the